

Roberto André Mossi Milla

Computer Vision Engineer | Machine Learning | AI Systems

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Summary

Computer Vision and Machine Learning Engineer building production AI systems spanning computer vision, multimodal AI, OCR, and full-stack deployment. Experience includes real-time YOLO detection, GPU-accelerated AI research, image processing, model retraining workflows, and AI systems that reduce previously manual processes from days or weeks to minutes.

Experience

Ingeniería Digital

May 2026 – Present

Software Engineer (AI Systems)

Honduras

- Built a multimodal AI vehicle-damage assessment platform that reduced assessment time from more than one week to approximately a minute, automatically generating damage analyses and insurance estimates from vehicle imagery and VIN information.
- Developed the end-to-end application across React, Next.js, TypeScript, PostgreSQL, AWS S3, and AI services, combining AWS Rekognition for vehicle localization, Gemini Flash for damage detection, VIN-history retrieval, and automated insurance estimation workflows.
- Designed an interactive AI evaluation workflow using Konva that allows users to inspect and correct detection regions, persist annotations, and evaluate different models, processing layers, and AI combinations in an experimental sandbox.
- Built Playwright-based data-acquisition services to retrieve and normalize vehicle information and VIN history from external sources for valuation and insurability analysis.
- Built an AI-assisted medical document generation platform from scratch, combining Amazon Textract OCR, OpenAI APIs, structured document generation, and cloud storage; reduced report preparation from approximately one week to a day, with final document generation completed in under one minute.
- Architected secure full-stack document workflows using React, Next.js, TypeScript, PostgreSQL, Auth0, and AWS S3 for document upload, AI processing, editing, versioning, and management.

Gannon University

Aug. 2024 – May 2025

Computer Vision Engineer — Industry-Sponsored Project

Erie, PA

- Engineered a real-time industrial computer-vision pipeline using LiDAR-triggered image capture and YOLOv8 to identify packages entering a warehouse and automatically record qualifying detection events for downstream damage analysis.
- Developed a live monitoring dashboard displaying detection bounding boxes and events, with administrative tools allowing operators to correct annotations and submit revised data for model retraining.
- Retrained YOLO models using curated datasets and improved image quality through contrast, color, filtering, spatial partitioning, and image-reconstruction techniques to support more reliable inference.
- Containerized distributed inference and data-processing services using Docker and integrated C++, Python, and Rust components for real-time system communication.
- Helped migrate the inference system from Raspberry Pi hardware after thermal and performance limitations were reached, supporting deployment on NVIDIA Jetson hardware while maintaining near-real-time visual feedback.
- Took primary ownership of the monitoring and visualization interface, integrating computer-vision events with TimescaleDB and backend services for operational review.

Gannon University

Aug. 2024 – May 2025

Research Assistant — Evolutionary Artificial Intelligence

Erie, PA

- Conducted undergraduate research on autonomous multi-agent learning and evolutionary computation, investigating emergent behavior in dynamic simulated environments using NeuroEvolution of Augmenting Topologies (NEAT).
- Independently implemented NEAT in C++ and designed a numerical genome representation encoding agent sensing, actions, and neural structure for evolutionary optimization across generations.
- Developed a procedurally generated 3D Unreal Engine environment capable of producing randomized or seeded halls, floors, walls, and objects for repeated multi-agent experiments, integrating C++, and CUDA.
- Contributed to the peer-reviewed paper “Tag AI-Sandbox,” published through the American Society for Engineering Education (ASEE) in 2025 and presented at a Sigma Xi conference at Penn State University.

Dinant

Jun. 2023 – Jul. 2023

Full-Stack Developer

Honduras

- Led end-to-end development of two production platforms used by two companies, replacing manual operational processes that previously required weeks with automated workflows completed in minutes; systems remain in use.
- Designed relational database schemas and full-stack workflows for contract management, digital signatures, automated PDF generation, repair tracking, machine maintenance, inventory deduction, and operational reporting.
- Collaborated directly with international stakeholders, gathering requirements, presenting iterative demonstrations, incorporating production feedback, and deploying finalized systems.

Open Source

Copilot-Lualine

2024 – Present

Creator & Maintainer

Open Source

- Created and maintain an open-source Neovim plugin providing interactive GitHub Copilot status indicators and controls for enabling, disabling, authentication, and logging; received 117+ GitHub stars and contributions from four community developers.

Technical Skills

Computer Vision & ML: YOLOv8, PyTorch, CUDA, Computer Vision, Multimodal AI, OCR, Image Analysis, Object Detection, Object Tracking, Reinforcement Learning, NEAT, LLMs

Languages: Python, C++, Rust, TypeScript, JavaScript, SQL, Java, Bash

Systems & Data: Docker, Linux, TimescaleDB, PostgreSQL, Redis, REST APIs, Event-Driven Services, AWS S3, Amazon Textract

Application Engineering: React, Next.js, Node.js, Playwright, Konva, Auth0, GitHub Actions, CI/CD

Education & Recognition

Gannon University

B.S. Computer Systems Engineering

- Peer-reviewed publication with ASEE; research presentation at Sigma Xi.

2025

Erie, PA